Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.00	20.03	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.00	2.00	+0.2%
power (W), mean	629	629	-0.1%
power (W), peak	652	664	+1.8%
energy (Wh)	210	210	+0.1%
energy, parked workers still on at idle power (Wh)	210	210	+0.1%
CPU used (cores), mean	0.893	0.880	-1.5%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.037	0.037	-1.5%
energy per core-hour (Wh)	705	715	+1.4%
node-hours per core-hour	6.72	6.82	+1.5%
pending pods, pod-minutes	0.267	0.267	+0.0%
pending pods, peak	1.00	1.00	+0.0%
HPA replicas, mean	9.02	9.01	-0.0%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.783	0.783	+0.0%
requests timed	1,035	1,150	+11.1%
time over the response line (% of samples)	0.290	0.000	-100.0%
response time (ms), mean	163	44.65	-72.7%
response time (ms), median	135	38.90	-71.1%
response time (ms), 95th percentile	321	66.66	-79.2%
response time (ms), 99th percentile	428	85.51	-80.0%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 9, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.1153	7.7231	+8.5%	[-1.3944, +2.6454]	EVEN, no significant difference
kwh_per_core_hour	0.7441	0.8045	+8.1%	[-0.1351, +0.2587]	EVEN, no significant difference
utilisation	0.0379	0.0363	-4.2%	[-0.0118, +0.0090]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.8466	0.7805	-7.8%	[-2.5397, +2.3415]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.4935	2.2685	-9.0%	[-4.4318, +4.3848]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 0.695
- heat (thermal) per minute: 0.358 0.391 0.42 0.444 0.467 0.486 0.502 0.518 0.531 0.541 0.55 0.556 0.562 0.567 0.572 0.577 0.58 0.583 0.585 0.586
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	616	4.75	4.00	0.000	0.000	0.545	0.512

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	620	615	5.00	6.00	0.000	0.000	0.625	0.499
2	6.00	6.00	620	616	5.50	6.00	0.000	0.000	0.646	0.527
3	6.00	6.00	622	620	6.00	6.00	0.250	0.250	0.706	0.666
4	6.00	6.00	632	636	10.00	8.75	0.000	0.000	0.970	1.08
5	6.00	6.00	632	633	10.00	10.00	0.000	0.000	0.974	0.977
6	6.00	6.00	633	635	10.00	10.00	0.000	0.000	0.984	1.02
7	6.00	6.00	646	654	10.00	10.00	0.000	0.000	1.34	1.58
8	6.00	6.00	649	646	10.00	10.00	0.000	0.000	1.40	1.35
9	6.00	6.00	648	651	10.00	10.00	0.000	0.000	1.42	1.47
10	6.00	6.00	641	648	10.00	10.00	0.000	0.000	1.19	1.39
11	6.00	6.00	616	615	10.00	10.00	0.000	0.000	0.562	0.504
12	6.00	6.00	619	617	10.00	10.00	0.000	0.000	0.623	0.527
13	6.00	6.00	625	622	10.00	10.00	0.000	0.000	0.789	0.691
14	6.00	6.00	634	636	10.00	10.00	0.000	0.000	1.00	1.08
15	6.00	6.00	637	632	10.00	10.00	0.000	0.000	1.10	0.963
16	6.00	6.00	632	632	10.00	10.00	0.000	0.000	0.973	0.979
17	6.00	6.00	627	626	10.00	10.00	0.000	0.000	0.844	0.810
18	6.00	6.00	620	616	10.00	10.00	0.000	0.000	0.643	0.521
19	6.00	6.00	619	615	10.00	10.00	0.000	0.000	0.591	0.536

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.